# Since we are talking about user experiments

Variety of ways to handle this:

- Analyse the data separately for differences in human characteristics
- For instance, if you think "age" is a significant factor in your experiment you might run an analysis to compare participants above/below some cutoff age
  - For example, children, older adults etc.

# Since we are talking about user experiments

Variety of ways to handle this:

- Deliberately specify such characteristics as random variables
- Typically, with suitable acknowledgement of possible limitations of this on your work
  - For instance, cultural norms differ in the East vs West — but often as researchers we just comment that our research does not look at this.
  - In HCI papers for example, you might see this in a limitation section at the end of a paper.

# Since we are talking about user experiments

You might have to change your questions based on participants:

- For example, rephrasing questions to be age appropriate if conducting an experiment with children, such as using the smileometer

# Within / Between Subject Designs / Counterbalancing

We need a to, structurally, control how we conduct an experiment to ensure a fair comparison

- For example, to mitigate things like ordering effects, learning, etc.

- The two most common in HCI are:
  - Within-subjects designs and between-subjects designs

# Within-Subjects Design

**Within-subjects** design:

- All subjects are exposed to **all conditions**;
  - i.e., they experience **every level** of the **independent variable(s)**;
  - Subjects are the things you are studying: people, images, search terms, etc.
- Also known as: **repeated-measures** design;
  - Because you repeat the measurements for every condition!

# Between-Subjects Design

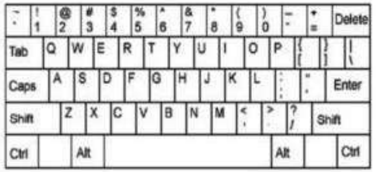
**Between-subjects** design:

- Subjects are placed into **groups** and exposed to just **one condition**;
  - i.e., they experience one level of the **independent variable(s)**;
- Also known as **between-groups** or **independent-measures** design;
  - Because you make independent measurements for each condition.

# Example:

Take the example comparing the two keyboard types:

QWERTY KEYBOARD




DVORAK Keyboard Layout


You have 20 willing participants for a study, a typing task and way of calculating typing speed and accuracy. You could approach the comparison in two ways:

**Within-subjects (repeated measures)**

All 20 participants use both keyboards and you compare the average difference in performance between the two.

**Between-subjects**

You split the participants into two groups of 10 and compare performance of QWERTY group to Dvorak group

# Within vs Between Subjects Design

Within-subjects experiments are **less common in other fields**;

- Example: in Medicine, the participants cannot take the treatment drug and the control drug at the same time or in sequence;
- **Between-subjects** designs are typically necessary in other fields because the subjects **can only be exposed to one** of the conditions.

This is usually not an issue in Computing Science;

- Our 'subjects' are often digital, so we can reuse their **initial state**;
  - E.g., we can take the same set of objects and give it to **multiple sort algorithms**;
- But there are exceptions: manipulating real-time data, studies with human participants, 'one-off' events, etc.

# Within vs Between Subjects Design

**Within-subjects** experiments are typically **fairer**;

- Because subjects are exposed to all conditions;
- Contrast with **between-subjects**, where you compare measurements from **different subjects** instead;

**Within-subjects** experiments are typically **cheaper**;

- If recruiting human participants, every person experiences **all *n* conditions**, rather than having to hire ***n* times as many people**;

# Within-Subjects Design

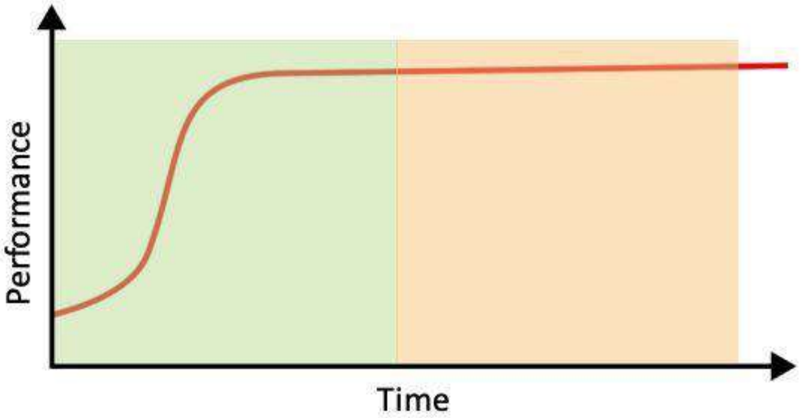
**Order effects** are a potential limitation to **within-subject** designs:

- When the **order of conditions** has a potential **impact on the DVs**;
- Most common in experiments with human participants;

**Learning effects** are a particular type of order effect:

- When participants gain knowledge and experience that benefits the conditions towards the end of the experiment;
- Example: participants achieve higher performance later in an experiment because they have **learned how the user interface works**;
- Example: participants perform a programming exercise better towards the end of the experiment because of **experience with earlier tasks**.

# Within-Subjects Order Effects



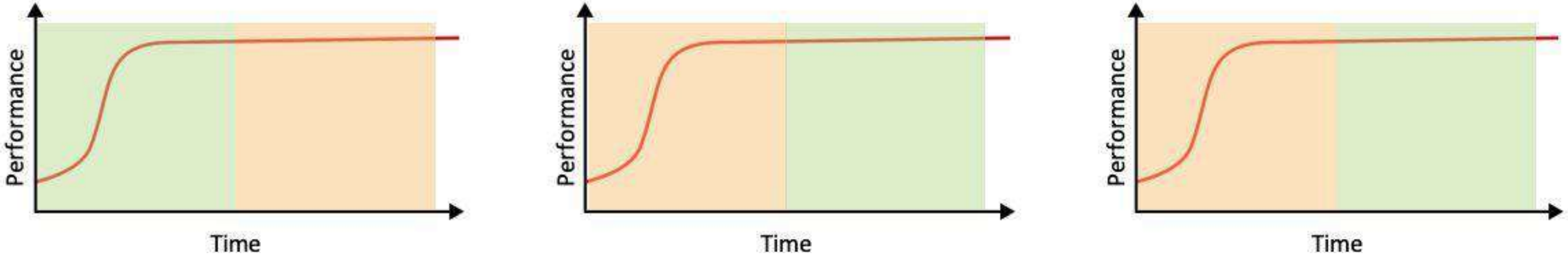

Learning occurs during the first condition, so performance will naturally be better during the second condition;

- The variance in our dependent variable is more likely to be caused by learning effects than differences in the independent variable(s).

# Within-Subjects Order Effects — Randomisation

**Randomising** condition order helps mitigate order effects:

- Conditions are presented in a random order for each participant;
- By randomising them, order effects get **distributed across conditions**;
  - In an attempt to 'cancel' the effects out.
- But is this optimal?
  - Depends too much on the randomiser…



# Counterbalancing — Latin Squares

**Counterbalanced** designs are an even better alternative:

- Instead of hoping the random order cancels out learning effects, counterbalanced designs are **systematically balanced**;

It would be impractical to use every unique condition ordering;

- For ***n*** *conditions* are ***n!*** orders;
  - E.g., four conditions means 24 condition orders.

Instead, use a **Latin Square**: an ***n*** *x* ***n*** array where each symbol occurs only once in each row and once in each column;

- A practical compromise: yields ***n*** condition orders such that each condition will appear just once in each position in the sequence.

# Counterbalancing — Latin Squares

| Order 1 | Order 2 | Order 3 | Order 4 |
|---|---|---|---|
| A | B | C | D |
| B | C | D | A |
| C | D | A | B |
| D | A | B | C |

**Latin Square**: an *n x n* array where each symbol occurs only once in each row and once in each column;

# Finally, it's important to know how to define an experiment

- Given a problem statement you should be able to propose an experimental design to answer it:

  Propose an experiment to determine which of two new  keyboards is "better"

# The experimental checklist

- Research Question
- Criteria
- Independent / Dependent Variables
- Conditions
- Tasks (or Processes)
- Experimental Objects
- Control Variables
- Random Variables
- Possible Confounding Factors

# First — what is better?

Better could mean any number of things:

- Latency of input
- Speed which a user can use it to type
- Accuracy which users can use it whilst typing
- User experience using the device
- Etc.

I’ll pick “user experience using the device” (which is still a big topic)

# Research question / criteria

- RQ (Stands for Research Question): Which of the two keyboards provides the better user experience?

- Here, we can define user experience to be:

    - Comfort whilst using the device (for long durations)
    - Workload whilst using the device

# We might even just explicitly say

RQ: Are there any significant differences in user comfort whilst using either keyboard over a long period of time (e.g. 1 week of continuous usage)?

RQ: Are there any significant differences between the user workloads whilst using either keyboard?

# Defining various variables / factors

Dependent variable:

- **Comfort** — questions measured on a 5-point Likert scale / qualitative feedback
- **Workload** — measured via a standardised questionnaire (e.g. the Nasa TLX questionnaire)

Note: We will cover more about capturing this data via questionnaires in the following weeks

# Defining various variables / factors

Independent variable: the two keyboards

**Task:** A recreation of office work — simulate 50 minutes office work (so various tasks such as writing emails, replying to slack messages, searching the web, writing a report, etc.)

**Conditions:** Keyboard One, Keyboard Two

# Defining various variables / factors

Experimental objects: the human participants

Control variables: the same task used for both conditions / all participants, the same room used for all participants, the same computer monitor / chair / etc. within the room

# Defining various variables / factors

**Random variables:** variations in human characteristics (e.g. prior technological experience)

**Confounding factors:** participants prior experience typing (e.g. touch typists vs single finger typers)

# Anything else?

- Still need to report how we intend to analyse our data (even at the experimental design stage)

- Still need some mention of the ethics of running the experiment (particularly if human participants are involved)
    - More on these topics next week

# Any Questions?

# Summary from this lecture

- The definitions of “Within/Between Subject Designs” and “Experimental Factors / Variables”
- Counterbalancing, randomisation
- Human characteristics and participaints
- Cofounding factors

- You need to know the definitions of all of these terms — these are the language used to describe / define experimental designs

# Quick Break Before Part 2